



Program and Branch Name:

[B.Tech CSE-F]

PBL Progress Report – I

Field	Details
Course Name	Data Structure and Algorithms - II
Course Code	CCSE0301
Faculty Name	Mr. Shamshad
Assignment Type	Individual Assignment
Student(s) Name	Divyanshu Pandey
ERP ID (I Year) / Roll No. (II Year onwards)	0251DCS037
Team/Group No. (Group/Capstone only)	Not Applicable — Individual Assignment
Problem Statement / Project Title	E-Cart
SDG(s)	SDG 9 – Industry, Innovation & Infrastructure
Reporting Month	Month 1
Date of Submission	07/09/2026

Student Declaration

I certify that this assignment is my own work, with all sources appropriately acknowledged. I understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature:

Date: 07/09/2026

Divyanshu Pandey

PBL Progress Report – I

Review 1: Problem Understanding & Initial Progress

Review Coverage:

- Individual Assignment: Unit/Module 1 (Trees) and Unit/Module 2 (Graphs)

Field	Details
1. Problem Statement / Assignment or Project Title	E-Cart – Online Shopping Management System is a web-based platform for browsing products, managing carts, placing orders and tracking deliveries. Administrators and vendors manage products, categories, inventory and orders.
2. Problem Context & Background	Manual attendance management is time-consuming, error-prone and difficult to search or update. The proposed system organizes student attendance efficiently using suitable data structures.
3. Problem Identification / Understanding	Managing many products and orders is difficult and time-consuming. Manual records may cause errors and inefficient searching. E-Cart organizes this information efficiently.
4. Objectives	Develop an organized, user-friendly online shopping platform for product search, cart, inventory and order management using DSA for efficient data organization and searching.
5. Target Users / Stakeholders	Customers browse products, manage carts, place orders and track deliveries. Vendors manage products, stock and orders. Administrators manage users, products, inventory and system operations.
6. Research / Literature Review Completed	Studied E-Cart and its major modules along with DSA-II Unit 1 (Trees) and Unit 2 (Graphs). Explored BST, Tree Traversal, Graphs, BFS and DFS.

7. Unit/Module Concepts Identified for Application	Time Complexity	DSA Concept	Possible Application	Purpose
	$O(\log n)$ avg.; $O(n)$ worst	Binary Search Tree (BST)	Organizing product records	Efficient product search
	$O(n)$	Tree Traversal (In-, Pre-, Post-order)	Traversing category/product data	Processing hierarchy
	$O(1)$ definition	Tree Terminology (Root, Level, Height, Subtree)	Category $\rightarrow$ Subcategory $\rightarrow$ Product	Understanding hierarchy
	Insert/Delete $O(\log n)$, Peek $O(1)$	Binary Heap	Priority-based products/operations	Managing priorities
	$O(V + E)$	Graph Representation	Customers, products, vendors, orders	Modeling connected entities
	Matrix $O(1)$, List $O(\text{degree})$	Adjacency Matrix / List	Storing E-Cart relationships	Structured access
	$O(V + E)$	BFS / DFS	Exploring connected E-Cart entities	Systematic traversal
	$O(1)$	Graph Terminology	Vertices and relationships as edges	Understanding connected data

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Review 1: Problem Understanding & Initial Progress

Field	Details
8. Work Completed During Month 1	Studied the E-Cart problem statement, requirements, modules and stakeholders. Analysed Trees and Graphs for project applications. Reviewed initial ER Diagram, DFD and database structure.
9. Evidence / Supporting Material	• SDLC phase documentation. • Initial system design concepts. • ER Diagram. • Data Flow Diagram (DFD). • Database table structures.
10. Challenges Faced	• Understanding DSA concepts in a real-world e-commerce system. • Selecting suitable data structures for E-Cart operations. • Organizing products by categories. • Representing hierarchy using Trees. • Representing relationships using Graphs.
11. Individual / Team Contribution	• Studied E-Cart problem statement and requirements. • Identified users, stakeholders and major functionalities. • Studied product, category, cart, order and inventory management. • Studied DSA-II Trees and Graphs and relevant concepts. • Mapped BST and Tree Traversal to product/category organization.
12. Overall Progress (%)	30% Initial problem understanding, requirements, objectives, stakeholder identification and DSA concept mapping are complete. Major E-Cart modules and initial system design were studied. Detailed implementation and testing will be carried out in later reviews.
13. Plan for Next Review	• Finalize system requirements and suitable data structures. • Apply Tree concepts to product/category organization. • Explore BST-based product searching. • Apply Graph concepts to E-Cart relationships.

Faculty Evaluation:

- To be completed by Faculty in Moodle